LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine air intake and propeller systems, windshield anti-icing systems, ice detection
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine air intake and propeller systems, windshield anti-icing systems, ice detection.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system schematics and circuit protection device ratings; Instructor supervision of student note-taking on actuator types.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the engine air intake system?
(b) Name two types of circuit protection devices used in avionics.
(c) Why is ice detection critical for propeller systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Engine air intake and propeller systems • Windshield anti-icing systems • Ice detection mechanisms	L & Ds	PPT, IB	15 min
(b)	• Circuit protection devices and generators • Actuators and engine starting system • System integration and safety	L & Ds	PPT, PDF	20 min
(c)	• Operational limits and maintenance checks • Troubleshooting common failures • Emergency procedures for icing	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the windshield anti-icing system prevents ice accumulation.
(b) Describe the role of circuit protection devices in the engine starting system.
(c) What are the indications of a malfunctioning ice detection system?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between engine air intake systems and propeller systems, referencing the next lesson plan on landing gear control.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Landing, gear control: - Landing gear position indication, anti-skid control systems, wind shield wiper systems, rain repellent systems, air conditioning systems.	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min